

Batch Sparge and Party Gyle Simulator

From German brewing and more

The bath sparging process is fairly predictable and can easily be modeled. This is done in the Batch Sparge and Party Gyle simulator (http://braukaiser.com/download/batch_sparge_simulator.xls), a spread sheet which allows planning of batch sparge and party gyle scenarios. Party Gyle is a style of brewing where the mash is lautered in batches. The resulting worts of different strengths may be blended or not to yield different beers from a single mash. In batch sparge brewing all these run-offs are collected in the same boil kettle to yield a single beer.

The process is simple. After all or part of the wort has been drained from the lautertun a batch of sparge water is added and mixed well with the grain. The goal is to dilute the wort that remained in the spent grain and distribute its extract evenly throughout the new wort volume. After that the wort is lautered and drained again. This may be repeated one more time but brewers rarely use more than 2 sparge batches which means collecting 3 run-offs.

The mathematical model simply calculates how much extract is transferred into the boil kettle with each run-off and how much extract remains in the lautertun before it is diluted by the addition of a known amount of water. The amount of extract that is dissolved during mashing can be calculated from the grist's extract potential and the Conversion Efficiency of that mash. It determines the strength of the first wort and to some extent the volume of the first wort.

The following article explains how to use the spreadsheet.

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Units

While the spreadsheet strictly uses metric units under the hood the user can specify which units should be used for weights, volumes, extract content and grain absorption. It makes sense to save a copy of the spreadsheet loaded with your unit preferences.

units	
weight (kg or lb)	kg
volume (l, qt or gal)	l
extract content (Plato or SG)	Plato
grain absorption (l/kg or gal/lb)	gal/lb

initial mash

To calculate the amount of extract that will be dissolved the amount of grain, its extract potential (default is 80% fine grind dry basis and 4% moisture content) and the conversion efficiency are needed. The conversion efficiency depends on your mashing parameters and you should get this number from previous mashes. See Conversion Efficiency on how to measure it. The water amount is needed to calculate the amount of resulting wort and its strength.

initial mash		
grain weight	4.5	kg
extract potential	80	%
moisture content	4	%
mash water	16	l
conversion efficiency	100	%

equipment

The only equipment parameters that batch sparging depends on are the specific grain absorption and dead spaces. For most brewers dead spaces are virtually nonexistent and commonly found grain absorption is 0.12 gal/lb or 1 l/kg which are the defaults here. If you have better data for your equipment, use it here. Note that the grain absorption and dead space volume is only used to calculate the amount of wort which can be drained. If you don't rely on that number but enter what you actually collected, these equipment parameters have no effect

equipment		
grain absorption (apparent)	0.12	gal/lb
dead spaces	0	l

1st run-off

Each run-off section gives the volume that can possibly be drained and its strength (gravity). The amount that can be drained is used as the default for the collected volume but can be changed. Based on the collected volume and the wort strength the efficiency collected with that run-off is calculated.

1st run-off		
possible volume	11.5	l
wort strength	17.8	Plato
collected volume	11.5	l
efficiency	63.2	%

boil off options

This section is particularly useful to get an idea how strong and how much wort is left after boiling. It lists the post boil volumes and strength for total, not hourly, evaporation percentages ranging from 5 to 30%. In general the evaporation should be between 10 and 15%, which is enough evaporation to drive off DMS and is not yet excessive. But larger beers may need or even benefit from a larger boil-off. The boil-off options are given for each run-off and all of the supported run-off combinations.

boil off options (%)						
	5	10	15	20	25	30
volume	10.9	10.3	9.7	9.2	8.6	8.0
strength	18.7	19.7	20.9	22.2	23.7	25.4
						l
						Plato

recharge

When batch sparging only water is added in preparation of an additional run-off. In party gyle brewing, however, the brewer may decide to add more grain and mash again to boost the gravity of the 2nd run-off.

1 st recharge		
water added	13	l
more grain added	0	kg
conversion efficiency	100	%

This is supported in the recharge sections where the amount of water and grain, which is added, can be entered. Along with the addition of grain is a conversion efficiency number which by default is set to whatever was used in the initial mash section.

combined run-offs

The spreadsheet supports up to 3 run-offs and all combinations with the exception of combining the 1st and the 3rd run-off. In a 2 run-off batch sparge, for example, you would look for the 1st + 2nd run-off section to find your total volume, wort strength and efficiency.

1st + 2nd run-off			
volume	24.5	Plato	%
wort strength	10.2		
efficiency	87.5		

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